

Order of Operations with Radicals

MA.8.NSO.1.7

I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

Yesterday the radical bar grouped whatever sat under it.

Today every grouping symbol at once — and the order comes from the structure, not from a mnemonic.

Warm-Up

On your own. Four minutes. No notes.

1. $\sqrt{5^2 - 9}$
2. How many significant digits are in 1.020 ?
3. Find the value: $\left(\frac{2}{3}\right)^{-2}$
4. Evaluate: $12 \div 3 \cdot 2$

Warm-Up

Answers.

1. $\sqrt{5^2 - 9}$ **4**

2. How many significant digits are in 1.020 ? **4**

3. Find the value: $\left(\frac{2}{3}\right)^{-2}$ **9/4**

4. Evaluate: $12 \div 3 \cdot 2$ **8**

Notes

Copy the table.

I. The structure, not a mnemonic

	Level	What it means
1	Grouping symbols	(), [], the radical bar, the fraction bar
2	Powers and roots	the same level as each other
3	Multiply and divide	left to right
4	Add and subtract	left to right

Innermost grouping symbol first, then out.

Notes

II. Left to right means left to right

$$12 \div 3 \cdot 2 = 4 \cdot 2 = 8, \quad \text{never} \quad 12 \div 6 = 2$$

$$10 - 4 + 3 = 6 + 3 = 9, \quad \text{never} \quad 10 - 7 = 3$$

Multiplication and division are one level. Addition and subtraction are one level. Left to right, both times.

Notes

III. The fraction bar groups, top and bottom

$$\frac{2^3 - 5}{6^2 - 27} = \frac{3}{9} = \frac{1}{3}$$

$$\frac{2^3 - 5}{6^2} - \sqrt{\frac{4}{9} - \frac{1}{3}} = \frac{1}{12} - \frac{1}{3} = -\frac{1}{4}$$

Finish the numerator and finish the denominator before you divide.

Example 1

A grouping symbol inside a grouping symbol.

$$4(8.63 - (3.2)^2) + \sqrt[3]{125}$$

Evaluate.

Example 1

Worked.

$$(3.2)^2 = 10.24 \quad \text{innermost: the power inside the parentheses}$$

$$8.63 - 10.24 = -1.61, \quad 4(-1.61) = -6.44 \quad \begin{array}{l} \text{finish the} \\ \text{parentheses,} \\ \text{then multiply} \end{array}$$

$$-6.44 + 5 = -1.44 \quad \text{the cube root is 5; add last}$$

Answer: -1.44

Your Turn

Same steps, your numbers. Boards up when done.

$$3(5.4 - (2.1)^2) + \sqrt{49}$$

Your Turn

Answer.

$$3(5.4 - (2.1)^2) + \sqrt{49}$$

$$3(0.99) + 7$$

Answer: 9.97

Example 2

A fraction bar and a radical in one expression.

$$\frac{2^3 - 5}{6^2} - \sqrt{\frac{4}{9} - \frac{1}{3}}$$

Evaluate.

Example 2

Worked.

$$\frac{2^3 - 5}{6^2} = \frac{3}{36} = \frac{1}{12} \quad \text{numerator and denominator, each finished}$$

$$\frac{4}{9} - \frac{1}{3} = \frac{1}{9}, \quad \sqrt{\frac{1}{9}} = \frac{1}{3} \quad \text{under the bar, then the root}$$

$$\frac{1}{12} - \frac{1}{3} = -\frac{1}{4} \quad \text{subtract last}$$

Answer: $-\frac{1}{4}$

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{3^2 - 1}{4^2} + \sqrt{\frac{7}{16} - \frac{3}{16}}$$

Your Turn

Answer.

$$\frac{3^2 - 1}{4^2} + \sqrt{\frac{7}{16} - \frac{3}{16}}$$

a half plus a half

Answer: 1

Whiteboards • Question 1 of 9

Boards up on three.

$$12 \div 3 \cdot 2$$

Answer it.

Left to right.

Whiteboards · Question 1 of 9

Answer.

$$12 \div 3 \cdot 2$$

$12 \div 3$ first

Answer: 8

Whiteboards • Question 2 of 9

Boards up on three.

A square tile has side $\sqrt{196}$ centimeters.

Its diagonal is about 19.8 centimeters.

What is the perimeter of the tile?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A square tile has side $\sqrt{196}$ centimeters.

Its diagonal is about 19.8 centimeters.

What is the perimeter of the tile?

4 sides of 14

Answer: 56 centimeters

Whiteboards • Question 3 of 9

Boards up on three.

$$10 - 4 + 3$$

Answer it.

Left to right.

Whiteboards • Question 3 of 9

Answer.

$$10 - 4 + 3$$

10 - 4 first

Answer: 9

Whiteboards • Question 4 of 9

Boards up on three.

$$\frac{2^3 - 5}{6^2 - 27}$$

Answer it.

Finish the top and the bottom.

Whiteboards • Question 4 of 9

Answer.

$$\frac{2^3 - 5}{6^2 - 27}$$

3 over 9

Answer: 1/3

Whiteboards • Question 5 of 9

Boards up on three.

$$2 + 3 \cdot 4^2$$

Which is the value?

A. 146

B. 50

C. 26

D. 80

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$2 + 3 \cdot 4^2$$

Which is the value?

A. 146

B. 50

C. 26

D. 80

Answer: B — 50

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{6^2 - 4}{2^3}$$

Which is the value?

A. 4

B. 35.5

C. 0.5

D. 256

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{6^2 - 4}{2^3}$$

Which is the value?

A. 4

B. 35.5

C. 0.5

D. 256

Answer: A — 4

Whiteboards • Question 7 of 9

Boards up on three.

$$-2^2 + (-2)^2$$

Answer it.

What is each exponent attached to?

Whiteboards · Question 7 of 9

Answer.

$$-2^2 + (-2)^2$$

-4 and +4

Answer: 0

Whiteboards • Question 8 of 9

Boards up on three.

$$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$$

Answer it.

Inside first.

Whiteboards • Question 8 of 9

Answer.

$$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$$

one eighth plus one ninth

Answer: 17/6

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Roscoe says that $(\sqrt{n} - 11)^2$ is always positive,

because a square is always positive.

Is he right? Give a value of n that settles it.

Write your answer in sentences.

Try to make the inside zero.

Whiteboards • Question 9 of 9

Answer.

Roscoe says that $(\sqrt{n} - 11)^2$ is always positive,
because a square is always positive.

Is he right? Give a value of n that settles it.

never negative is not the same as always positive

Answer: No — at $n = 121$ the value is 0

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate radical expressions — D8J